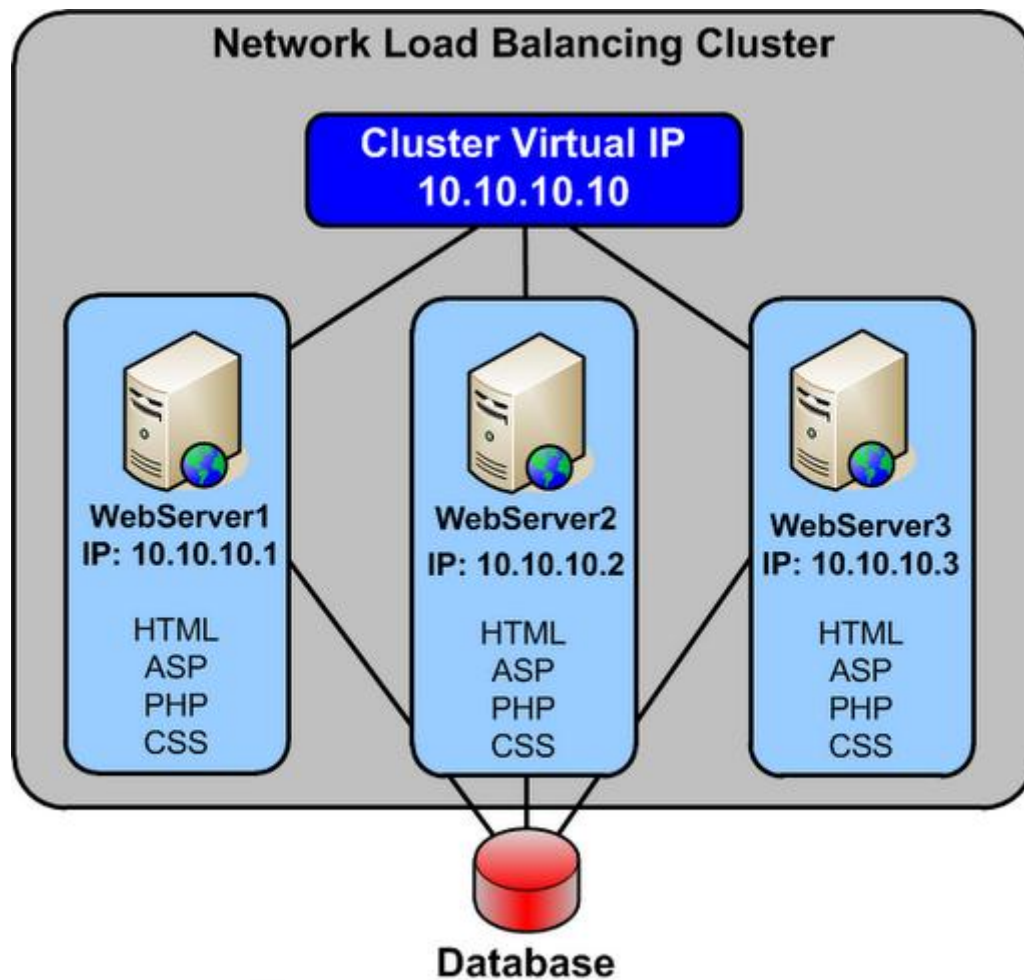


Install and configure NLB cluster in server 2019-09

1. Introduction

Network Load Balancing (NLB) in Windows Server 2019 is used to distribute client requests across multiple servers. This increases availability and scalability of services like websites, applications, or remote access.



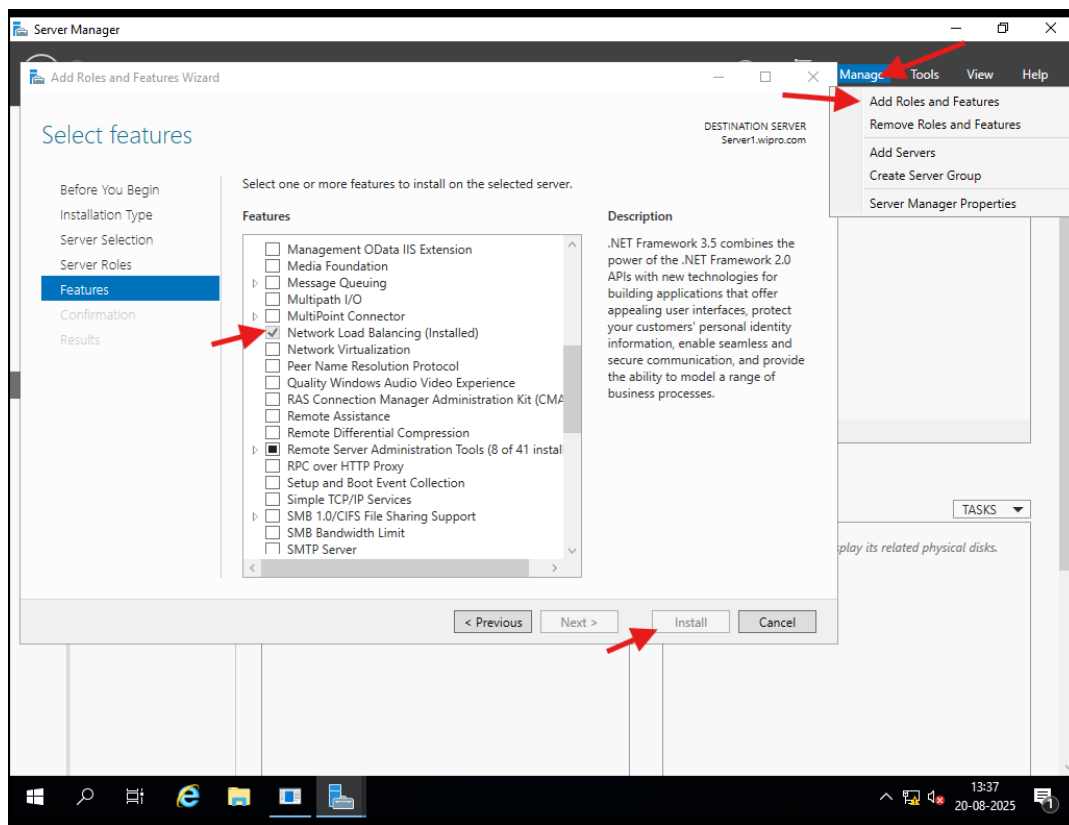
2. Prerequisites

- Minimum **two Windows Server 2019 systems** (physical or virtual).
- Both servers must have **static IP addresses**.
- Administrative rights on both servers.
- Same application or service (e.g., IIS) installed on all nodes.

3. Steps to Install and Configure NLB

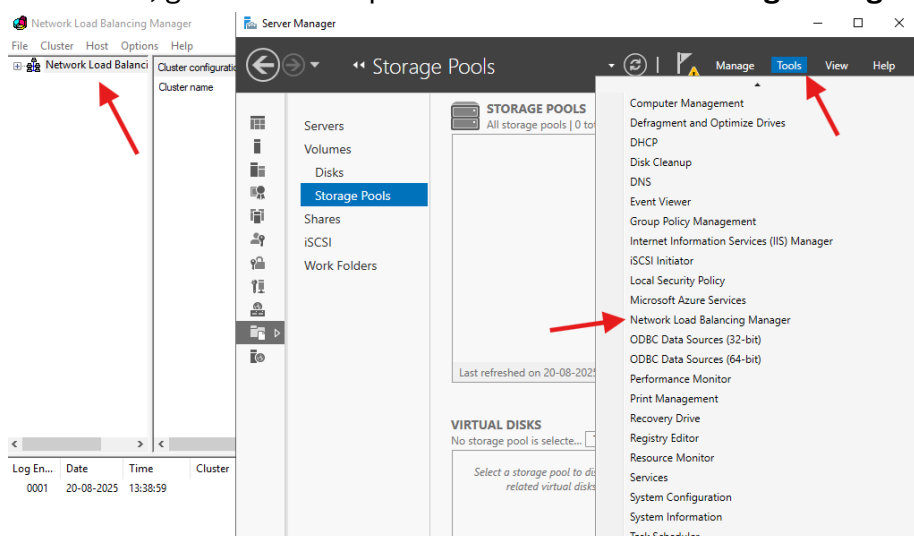
Step 1: Install the NLB Feature

- On each server, open **Server Manager** → **Manage** → **Add Roles and Features**.
- Select **Network Load Balancing** feature.
- Complete installation on all cluster nodes.



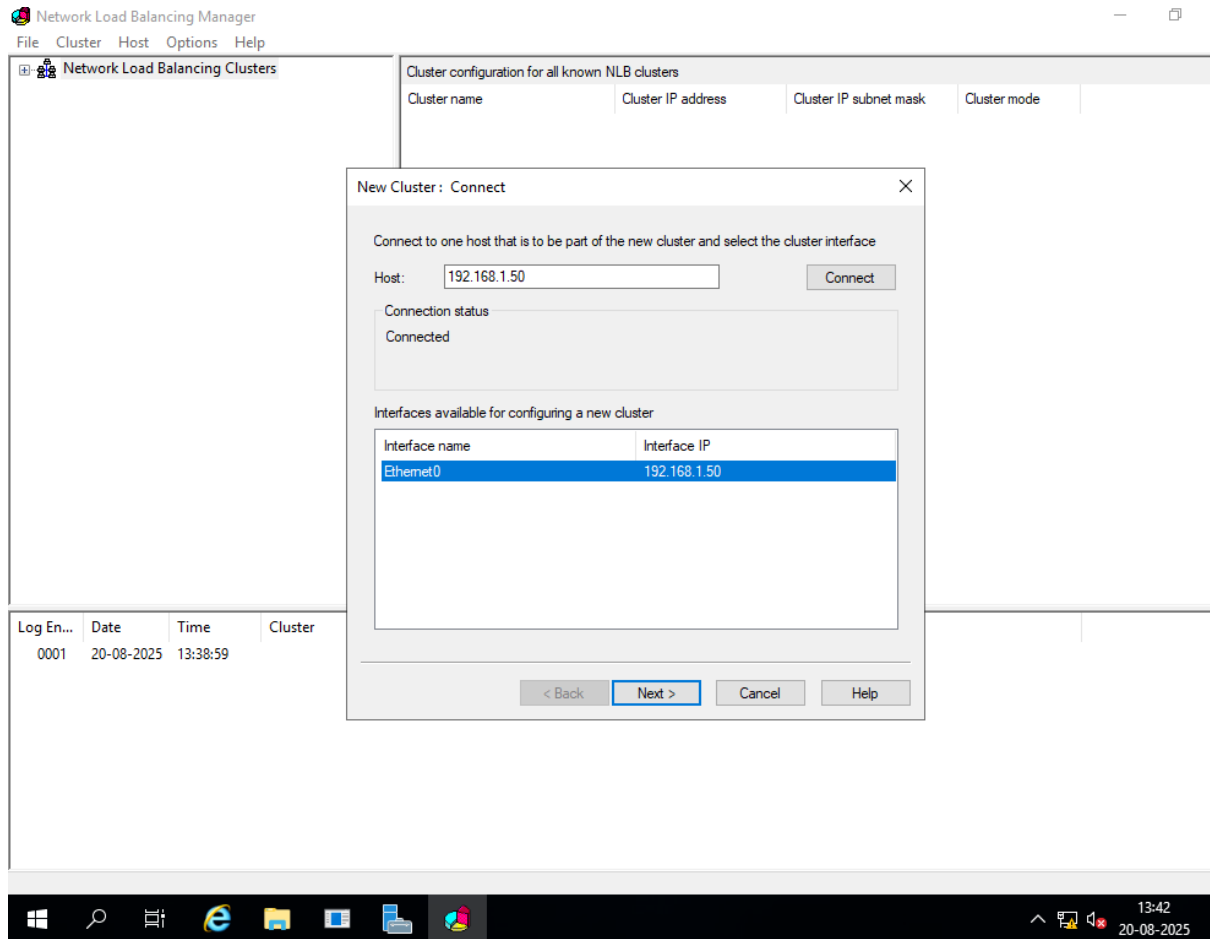
Step 2: Open NLB Manager

- one server, go to **tools** → Open **Network Load Balancing Manager**.



Step 3: Create New Cluster

- In NLB Manager, right-click → **New Cluster**.
- Enter the first server's IP address → Connect → Select the network interface.



- Configure **host parameters** (priority, dedicated IP).
- Click **Next**.

Step 4: Configure Cluster IP Address

- Add a **Cluster IP** (this will be the shared virtual IP for clients). Example: 192.168.1.200.

The cluster IP addresses are shared by every member of the cluster for load balancing. The first IP address is the cluster IP address.

Cluster IP configuration

☒ Add IPv4 address:

IPv4 address: 192 . 168 . 1 . 200

Subnet mask: 255 . 255 . 255 . 0

☐ Add IPv6 address:

IPv6 address:

☐ Generate IPv6 addresses:

☐ Link-local ☐ Site-local ☐ Global

OK Cancel

< Back Next > Cancel Help

- Assign the cluster **Full Internet Name** (e.g., cluster.lab.local).

Cluster IP configuration

IP address: 192.168.1.200

Subnet mask: 255 . 255 . 255 . 0

Full Internet name: cluster.lab.local

Network address: 02-bf-c0-a8-01-c8

Cluster operation mode

☒ Unicast

☐ Multicast

☐ IGMP multicast

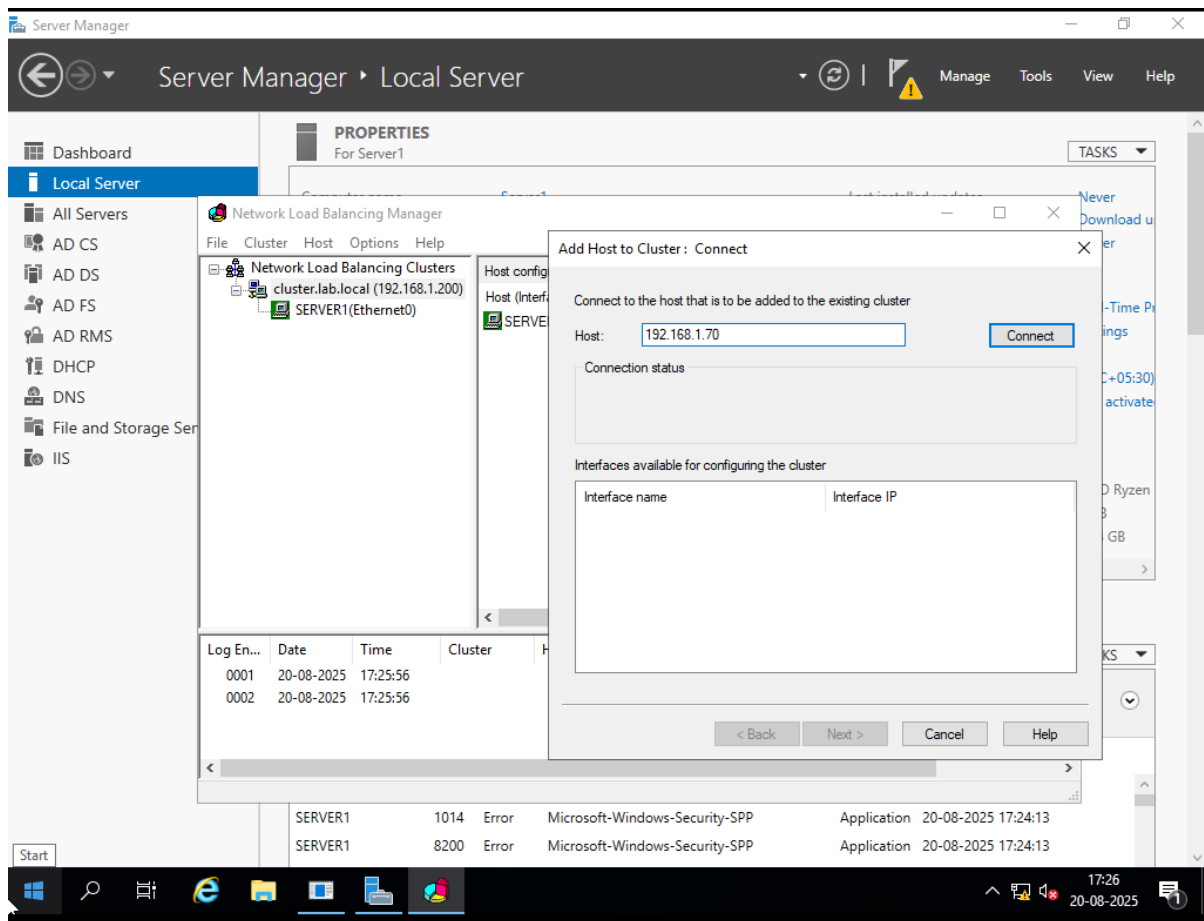
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- Choose **Unicast** or **Multicast** cluster operation mode.

Step 5: Define Port Rules

- By default, NLB adds a rule for all ports (0-65535).
- You can modify port rules if you only want specific ports (e.g., 80 for web, 443 for SSL).

Step 6: Add Additional Hosts



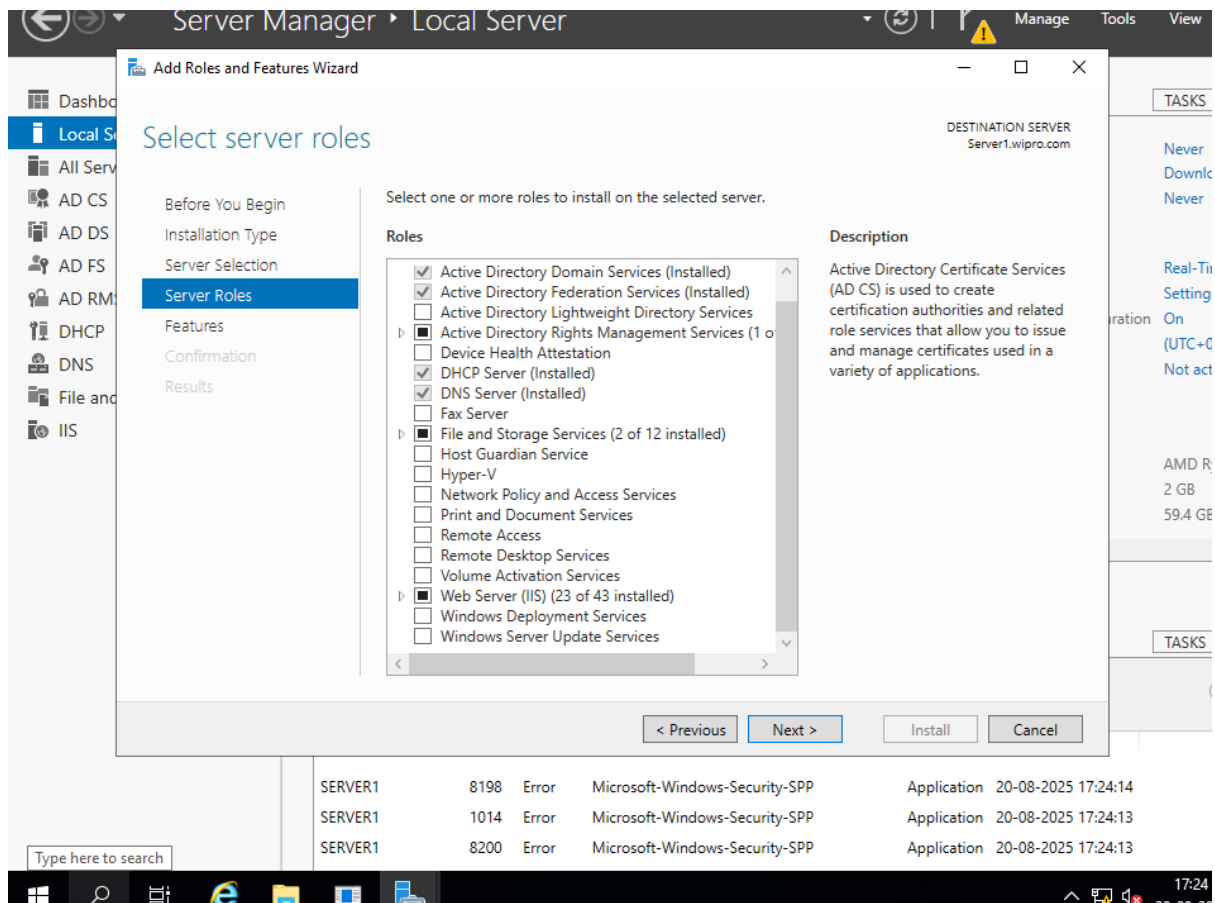
- After cluster creation, right-click the cluster → **Add Host to Cluster**.
- Enter the second server's IP address.
- Configure host priority and dedicated IP.
- Finish wizard.

Step 7: Verify Cluster Status

- Both servers should show **Converged** in NLB Manager.
- The cluster is now ready.

4. Testing the NLB Cluster

1. Install **IIS Web Server** on both nodes.



2. Place different index.html files on both servers (e.g., Server1: "This is Node1", Server2: "This is Node2").
3. Access the cluster IP (<http://192.168.1.200>).
4. Refresh multiple times; traffic should load balance between both servers.